

ORIGINAL ARTICLE

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Novel computer vision algorithm for accurate multi-label classification of concrete bridge defects

Jamiu Lateef¹ and Xiong Yu^{1*}

Abstract

Maintaining the structural integrity of reinforced concrete bridges necessitates the timely and accurate detection of surface defects. Conventional inspection methodologies remain labor-intensive, inherently subjective, and susceptible to human error, driving the need for automated assessment frameworks. This study introduces a multi-label defect classification model tailored for reinforced concrete bridge inspection, engineered to process imagery consistent with prevailing bridge inspection standards. The proposed framework is designed to simultaneously identify multiple co-occurring defects within a single image, addressing the practical reality of overlapping deterioration mechanisms. Leveraging the open-source Concrete Defect Bridge Image Dataset (CODEBRIM), three distinct ImageNet-pretrained deep neural network architectures were subjected to systematic hyperparameter optimization and fine-tuning to enhance classification performance across bridge-relevant defect categories. Beyond achieving high per-class accuracy, the optimized model attained a subset accuracy of 84.0% and a micro-averaged F1-score of 85.2% on a held-out test set, signifying robust recognition of overlapping distress conditions. Furthermore, evaluation on a synthetically generated dataset validated the model's generalization capacity under domain shift. The findings demonstrate that the proposed framework effectively supports automated defect documentation and holds significant potential for enhancing the objectivity and efficiency of bridge condition assessment protocols.

Keywords Multi-label defect classification, Deep neural network, Concrete structural defects, Bridge inspection

1 Introduction

The timely identification and remediation of defects in concrete structures are essential prerequisites for ensuring long-term structural safety and serviceability. Accurate detection of distress manifestations, such as cracks and spalling, facilitates proactive maintenance intervention and mitigates the risk of catastrophic structural failure (Deepanraj & Mewada, 2024). Within the domain of bridge asset management, defect identification

constitutes a foundational step in evaluating structural condition and informing subsequent maintenance prioritization. In the United States, visual inspection remains the predominant methodology for assessing bridge condition. Industry practice typically involves inspectors documenting the extent and severity of deficiencies on individual bridge elements and subsequently updating inspection records (Adhikari et al., 2014). This established workflow necessitates close-range access to structural surfaces and frequently relies on manual measurement using handheld instrumentation (Zakaria et al., 2022). Although effective in numerous contexts, visual inspection is characterized by significant temporal and financial costs, and outcomes may exhibit variability attributable to subjective judgment and inherent measurement uncertainty (Al-Sabbag et al., 2022). Access

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constraints in difficult-to-reach locations further compound operational difficulties and elevate safety risks.

Recent technological advancements in civil infrastructure monitoring have created unprecedented opportunities to enhance inspection efficiency and standardize defect documentation practices. Unmanned Aerial Vehicles (UAVs), for instance, have emerged as increasingly practical and economical platforms for bridge inspection, capable of rapidly acquiring large volumes of high-resolution imagery. However, translating these image repositories into actionable damage assessments remains a labor-intensive endeavor, as the interpretation and annotation of defects still demand substantial manual review (Silva & Lucena, 2018). Consequently, Computer Vision (CV) methodologies have surfaced as a promising avenue for automating defect identification and facilitating scalable, data-driven inspection workflows.

Traditional machine learning methods rely on hand-crafted features, such as edge-based and texture-based descriptors, which exhibit pronounced sensitivity to variations in illumination, surface texture, and imaging conditions. In contrast, deep learning models, particularly Convolutional Neural Networks (CNNs), possess the capacity to autonomously learn hierarchical feature representations directly from raw data and have demonstrated state-of-the-art performance across a diverse spectrum of CV tasks, including object detection, semantic segmentation, and image classification (Krizhevsky et al., 2017; LeCun et al., 1998). In civil infrastructure applications, CNN-based investigations have predominantly formulated defect recognition as a single-label classification problem. For instance, CNNs have been demonstrated to surpass classical edge-detector baselines for image-based concrete crack detection (Dorafshan et al., 2018). Similarly, research has addressed pavement crack detection (Zhang et al., 2016), and transfer-learning frameworks have been proposed to leverage multi-source knowledge for improved crack identification (Yang et al., 2020). While these studies underscore the efficacy of deep learning for defect recognition, their single-label formulation fails to adequately reflect field inspection conditions wherein multiple distinct distress types frequently manifest concurrently within a single image.

Multi-label image classification directly addresses this practical setting by predicting a set of labels that jointly characterize a given image (Tsoumakas & Katakis, 2007). Although multi-label classification has been extensively investigated in alternate domains (Khan et al., 2020; Tsoumakas & Katakis, 2007), concrete defect classification research has largely remained confined to single-label paradigms (Silva & Lucena, 2018; Deepanraj & Mewada, 2024). This emphasis leaves a critical gap for bridge inspection workflows, wherein defect

co-occurrence is prevalent and the cardinality of present defects varies substantially across images. For example, cracks may co-occur with efflorescence and corrosion stains; a practical classifier must therefore accurately capture these combinatorial label sets to support the generation of consistent and comprehensive digital inspection records.

To address this identified gap, the present study introduces a robust framework for multi-label classification of concrete bridge defects utilizing the Concrete Defect Bridge Image (CODEBRIM) dataset (Mundt et al., 2019) in conjunction with ImageNet-pretrained CNN backbones. The proposed framework is specifically designed to process inspection imagery wherein multiple distress types may co-occur. A systematic comparison is conducted between hyperparameter-optimized frozen-backbone transfer learning and fine-tuned transfer learning strategies, employing a suite of complementary multi-label evaluation metrics, including threshold-dependent measures and ranking-based evaluation. Furthermore, this study augments the evaluation protocol through the generation of a synthetic defect dataset using StyleGAN2-ADA, thereby enabling a rigorous assessment of model robustness under domain shift. The proposed approach not only advances the state of multi-label defect recognition but also establishes a comprehensive paradigm for evaluating model performance in this context.

2 Methodology

2.1 Dataset description

Publicly available bridge-inspection datasets are scarce, as the acquisition of field imagery, precise defect identification, and generation of high-quality annotations necessitate specialized domain expertise and substantial resource allocation. Accordingly, all models developed within this investigation were evaluated using the CODEBRIM dataset (Mundt et al., 2019). Designed to reflect the realities of in-service structures, the CODEBRIM dataset was assembled to capture overlapping and co-occurring defects under diverse acquisition conditions, allowing data-driven models to be trained and tested in settings that resemble practice.

The dataset comprises images collected from 30 distinct bridges exhibiting varying degrees of deterioration and structural damage. Photographs were acquired across a spectrum of environmental and structural contexts, encompassing variable illumination, viewpoints, and surface characteristics, thereby promoting the robustness and generalizability of models trained upon this data. CODEBRIM provides image-level annotations for six discrete categories: Crack, Spalling, Exposed Bars, Efflorescence, Corrosion Stain, and a non-defective Background class. The class-wise

frequencies are distributed as follows: Crack (2,507), Spalling (1,898), Corrosion Stain (1,559), Efflorescence (833), Exposed Bars (1,507), and Background (2,506). Each image is paired with a CSV record that lists all applicable labels, thereby permitting the association of multiple defect types with a single image. Figure 1 summarizes the distribution of images across the defined classes.

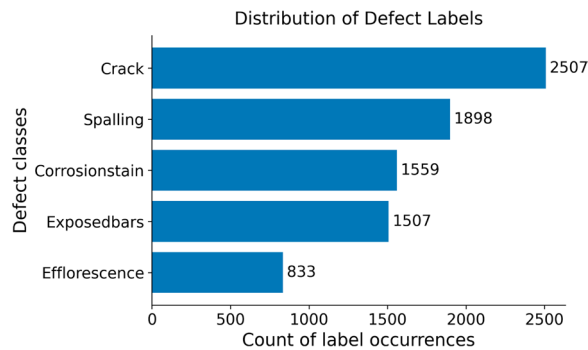


Fig. 1 Per-class image counts in the CODEBRIM dataset (Mundt et al., 2019)

Furthermore, the dataset contains a subset of images characterized by poor quality or low resolution. These images were systematically identified and excised through a combination of manual visual inspection and the application of the Laplacian variance image processing technique for blur detection (Bansal et al., 2017). Images whose Laplacian variance fell below a predefined threshold of 50 were flagged and subsequently removed from the dataset. Figure 2a–c provide illustrative examples of the low-quality and poor-resolution images that were eliminated during this preprocessing phase. Consequently, the final curated dataset utilized for model development consisted of 7,729 high-quality images. Additionally, the dataset contains images of varying sizes. Since CNNs require fixed-size images, all images are resized to 128×128 pixels to standardize input dimensions across models and to keep training and hyperparameter tuning computationally tractable on the available GPU resources. This resolution enables multiple transfer-learning experiments within realistic memory and runtime constraints. The curated dataset was subsequently partitioned into training, validation, and test subsets according to a 70%:20%:10% stratified split ratio.

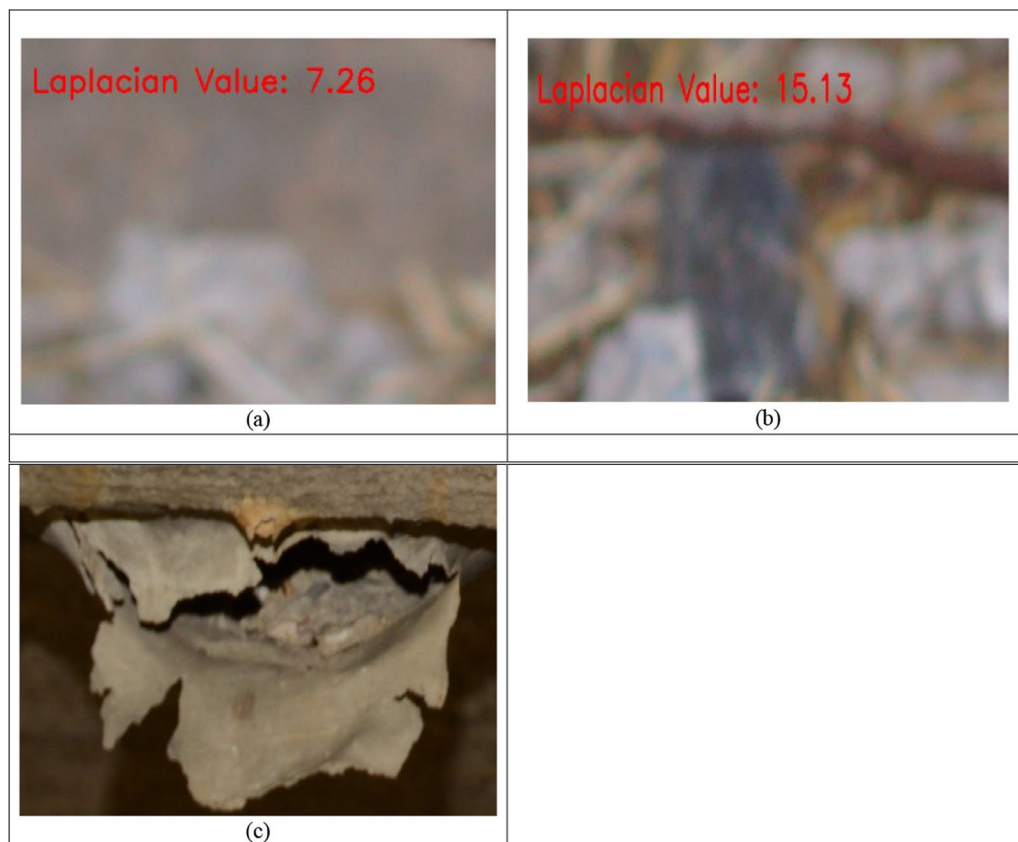


Fig. 2 Examples of poor and low-resolution images in the dataset before preprocessing. **a** blurring based on Laplacian variance, **b** blurring based on Laplacian variance, **c** unbalanced brightness based on visual judgment

2.2 Model implementation

This section delineates the implementation specifics for the multi-label classification task. All experiments were conducted utilizing a transfer learning paradigm, employing ImageNet-pretrained backbones augmented with a shared, task-specific multi-label classifier head. Input images were resized to 128×128 pixels in RGB format and normalized using the established ImageNet channel statistics (mean and standard deviation). The common classifier head architecture comprises a Dense layer with Rectified Linear Unit (ReLU) activation, followed by Batch Normalization, Dropout regularization, and a terminal Dense layer equipped with six Sigmoid activation units to facilitate independent probability estimation for each defect class.

Two training configurations were systematically evaluated: (i) Pretrained (frozen), in which the backbone was kept fixed and only the classifier head was trained; and (ii) Fine-tuned, in which the head and a subset of the final backbone stages were unfrozen and optimized with a lower learning rate. The number of unfrozen stages, along with the learning rate, optimizer, and dropout rate, was selected using Bayesian Optimization. Training was conducted with binary cross-entropy loss, early stopping, and a reduce-on-plateau learning-rate scheduler. A diagrammatic illustration of the proposed methodological framework is presented in Fig. 3. Comprehensive architectural details and the hyperparameter optimization protocol are elaborated in Sects. 2.2.1 and 2.2.2, respectively.

Training was carried out on Google Cloud Pro, provisioned with 83 GB of system RAM, 40 GB of GPU memory, and 113 GB of disk space. The environment used an NVIDIA A100-SXM4-40 GB (CUDA 12.4, driver 550.54.15), which supported running experiments across multiple models with varied learning rates and epoch counts.

2.2.1 Model architectures

Three ImageNet-pretrained backbone architectures were evaluated, selected to represent a spectrum of architectural paradigms, spanning classic convolutional stacks, residual and densely connected networks, and architectures derived via neural architecture search.

2.2.1.1 Visual group geometry-16 (VGG16) The VGG16 model is a deep convolutional neural network developed for large-scale image recognition, widely adopted as a robust feature extraction backbone owing to its architectural simplicity and consistent baseline performance. The network comprises 16 weight layers, including 13 convolutional layers employing 3×3 filters and 2×2 max-

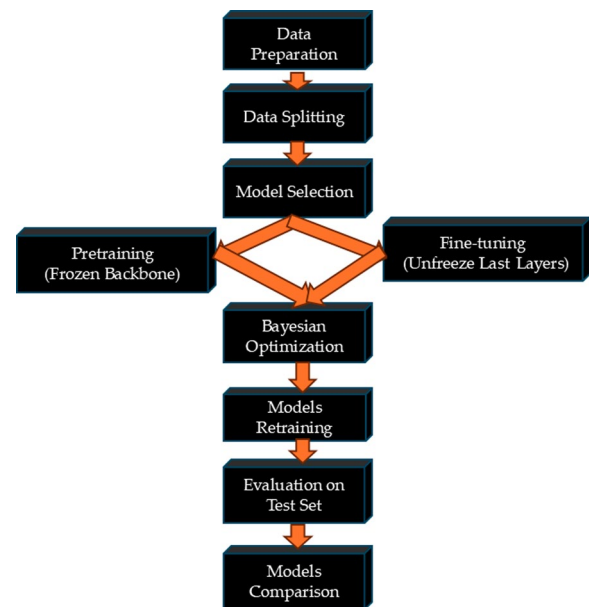


Fig. 3 Diagrammatic illustration of the proposed method

pooling layers arranged in sequence to extract hierarchical features. ReLU activation functions introduce non-linearity and enhance the model's learning capacity. The convolutional stack is succeeded by three fully connected layers, which account for the majority of the network's trainable parameters and are instrumental in synthesizing the extracted features for final classification. The architecture culminates in a Softmax activation function to yield class probability distributions.

2.2.1.2 ResNet50 The ResNet50 model is another popular deep convolutional neural network, known for its innovative use of residual learning (He et al., 2016). The network consists of 50 layers organized into a series of residual blocks that address the vanishing gradient problem and enable the training of deeper networks (Albahli & Yar, 2022). Each residual block integrates identity mappings and shortcut connections, enabling unimpeded gradient flow and thereby enhancing both training efficiency and accuracy. Within the architecture, 3×3 convolutional filters are combined with batch normalization and ReLU activation functions to extract meaningful features at multiple levels of abstraction. The integration of global average pooling preceding the fully connected layers enables ResNet50 to consolidate feature representations efficiently. A Softmax function is ultimately applied to derive class-specific probabilities.

2.2.1.3 EfficientNetB0 EfficientNetB0 is the baseline model within the EfficientNet family, engineered to optimize the balance between predictive accuracy and com-

computational efficiency through the principle of compound scaling. The architecture employs a systematic compound scaling methodology that uniformly scales network depth, width, and input resolution. EfficientNetB0 is constructed from multiple Mobile Inverted Bottleneck Convolution (MBCov) units, which leverage depthwise separable convolutions to reduce computational overhead without compromising feature extraction fidelity. Each block incorporates Squeeze-and-Excitation (SE) mechanisms to adaptively recalibrate channel-wise feature responses. The architecture utilizes batch normalization and the Swish activation function to promote training stability. EfficientNetB0 concludes with global average pooling and a fully connected classification layer, achieving an exceptional tradeoff between model compactness and predictive accuracy.

2.2.2 Hyperparameter optimization

Bayesian Optimization (BO) was employed to identify high-performing and directly comparable training configurations for each candidate backbone. Hyperparameter search procedures were implemented utilizing the Keras Tuner framework, with the optimization objective defined as minimization of validation loss computed via binary cross-entropy. To ensure equitable comparison, the hyperparameter search space and all fixed training parameters were maintained uniformly across VGG16, ResNet50, and EfficientNetB0 architectures. The shared search space and fixed settings are delineated in Table 1.

To avoid architectural selection bias, separate BO searches for each architecture were conducted, using the same search space and training settings, with 60 trials per architecture. During tuning, each trial was trained for 50 epochs with early stopping (patience=10) and ReduceLROnPlateau (factor=0.1, patience=5, min LR=1e-7). For each architecture, the top 10 configurations, ranked by validation loss, Accuracy, and F1 score on the validation set, were rebuilt and retrained for up to 200 epochs under the same settings. The singular best checkpoint for each architecture, selected based on validation performance, was advanced for final evaluation on the held-out test set. This two-stage, architecture-balanced strategy constrains hyperparameter confounding and facilitates a rigorous, direct comparison of baseline models. The model-specific optimal hyperparameter values identified through this procedure are reported in Table 2.

2.3 Evaluation metrics

Each retrained model was evaluated on the held-out test set using the following metrics: Accuracy, Precision, Recall, F1-score, Hamming Loss, Label ranking average precision (LRAP), and Coverage Error. Test set accuracy

Table 1 Search spaces and fixed training settings

Item	Value
Dense layer units	512, 1024, 2048, 4096
No. of unfrozen last layers	1, 2, 3, 4, 5
Dropout rates	0.3, 0.4, 0.5, 0.6
Kernel regularizer	0.0, 1e-4, 1e-3, 1e-2
Batch normalization	True, false
Optimizers	Adam, RMSprop, SGD
Learning rates	1e-3, 5e-4, 1e-4, 5e-5, 1e-5
Batch size	100
Learning rate scheduler	ReduceLROnPlateau (factor of 0.1, patience=5)
Metric	Validation loss, accuracy, and F1 score
Loss function	Binary cross-entropy
Image input size	128 × 128
Output activation	Sigmoid
Trial	60
Initial epoch (tuning)	50
Final epoch (retraining)	200

is reported as subset accuracy (also termed exact match ratio), wherein a prediction is counted as correct only if all labels for an image are predicted correctly. Precision, Recall, and F1-score are computed over the full test set across all classes. Hamming Loss, LRAP, and Coverage Error are computed on the full multi-label prediction matrix. The equations are given below:

$$\text{Accuracy}_{\text{subset}} = \frac{1}{N} \sum_{i=1}^N I(\hat{Y}_i = Y_i) \quad (1)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

$$\text{Hamming Loss} = \frac{1}{N \cdot L} \sum_{i=1}^N \sum_{\ell=1}^L \mathbb{I}[y_{i,\ell}^{\text{true}} \neq \hat{y}_{i,\ell}^{\text{pred}}] \quad (5)$$

$$\text{LRAP} = \frac{1}{N} \sum_{i=1}^N \frac{1}{|Y_i|} \sum_{y \in Y_i} \frac{|\{k \in Y_i : s_{ik} \geq s_{iy}\}|}{|\{k : s_{ik} \geq s_{iy}\}|} \quad (6)$$

$$\text{Coverage Error} = \frac{1}{N} \sum_{i=1}^N \max_{y \in Y_i} \text{rank}_i(y) \quad (7)$$

Table 2 Best hyperparameters by model

Model	No. of unfrozen last layers	KR	BN	Dense			Dropout rate			Optimizer	LR
				Unit 1	Unit 2	Unit 3	Unit 1	Unit 2	Unit 3		
VGG16	3	0.01	True	4096	2048	1024	0.5	0.5	0.5	Adam	$5e^{-5}$
ResNet50	2	0.01	True	2048	4096	1024	0.4	0.5	0.6	SGD	$1e^{-5}$
EfficientNetB0	3	0.01	True	4096	2048	1024	0.3	0.5	0.5	Adam	$1e^{-5}$

KR Kernel regularizer, BN Batch Normalization, LR Learning rate

In addition to aggregate test-set performance, models were further evaluated using class-wise metrics. For each defect label c , performance was quantified by treating the task as a one-vs-rest binary classification problem over the test set. Class-wise accuracy, precision, recall, and F1-score were computed using the standard binary definitions below, where TP_c , TN_c , FP_c , and FN_c are computed separately for each class c :

$$\text{Accuracy}_c = \frac{TP_c + TN_c}{TP_c + TN_c + FP_c + FN_c} \quad (8)$$

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \quad (9)$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \quad (10)$$

$$F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (11)$$

where N denotes the number of test images, and L represents the number of labels. Y_i and $\hat{Y}_i$ are the ground-truth and predicted binary label vectors for image i , respectively. Y_i is the set of positive labels for image i , $s_{i,y}$ is the predicted score for label y , and $\text{rank}_i(y)$ returns the rank position of label y when labels are sorted in descending order of predicted score.

2.4 Synthetic image generation

To further assess how each model performs on a new, distinct dataset, StyleGAN2-ADA (Karras et al., 2021) was used to synthesize images for all five defect categories and the background class. For each category, CODEBRIM defect images were gathered and resized to a uniform resolution of 512×512 for StyleGAN2-ADA training. This resolution was selected to preserve fine-grained defect characteristics while keeping GPU memory usage and training time at a practical level. The resulting class-specific training sets reflected realistic, concrete surface textures, illumination variations, and

surrounding scene conditions, providing suitable inputs for learning representative visual patterns.

A discrete StyleGAN2-ADA model was trained independently for each defect class, with adaptive discriminator augmentation enabled to enhance training stability on the comparatively limited per-class datasets and to foster the generation of diverse yet photorealistic samples. Following training, each generator was employed to produce synthetic images corresponding to its designated target class. The generated outputs were subjected to manual review to eliminate images exhibiting overt artifacts or ambiguous defect representations. Subsequently, the Fréchet Inception Distance (FID) was computed between the real and synthetic image distributions for each class (Heusel et al., 2017). Following prior best practices (Zhao et al., 2021), only synthetic images yielding an $FID < 50$ were retained for subsequent evaluation. This two-stage quality control protocol ensured that only high-fidelity synthetic images were utilized, thereby mitigating the risk that unrealistic artifacts would confound performance assessment. Representative examples of the generated synthetic images are presented in Fig. 4.

3 Results

This section presents a comparative evaluation of all models on the held-out test set. Both frozen-pretrained backbone variants and fine-tuned configurations are analyzed utilizing Subset Accuracy, F1-score, Hamming Loss, LRAP, and Coverage Error, supplemented by per-class performance analysis and Grad-CAM visualizations.

3.1 Pretrained (frozen backbone) results

This subsection reports test-set performance for the pre-trained, frozen-backbone configuration, in which the ImageNet-pretrained feature extractor is held fixed and only the multi-label classification head is optimised. As shown in Table 3, the three architectures exhibit comparatively narrow performance margins, suggesting that while generic ImageNet representations provide a viable foundation for transfer, they are insufficient to fully encapsulate the nuanced visual signatures of bridge defects and their inherent label co-occurrence patterns.

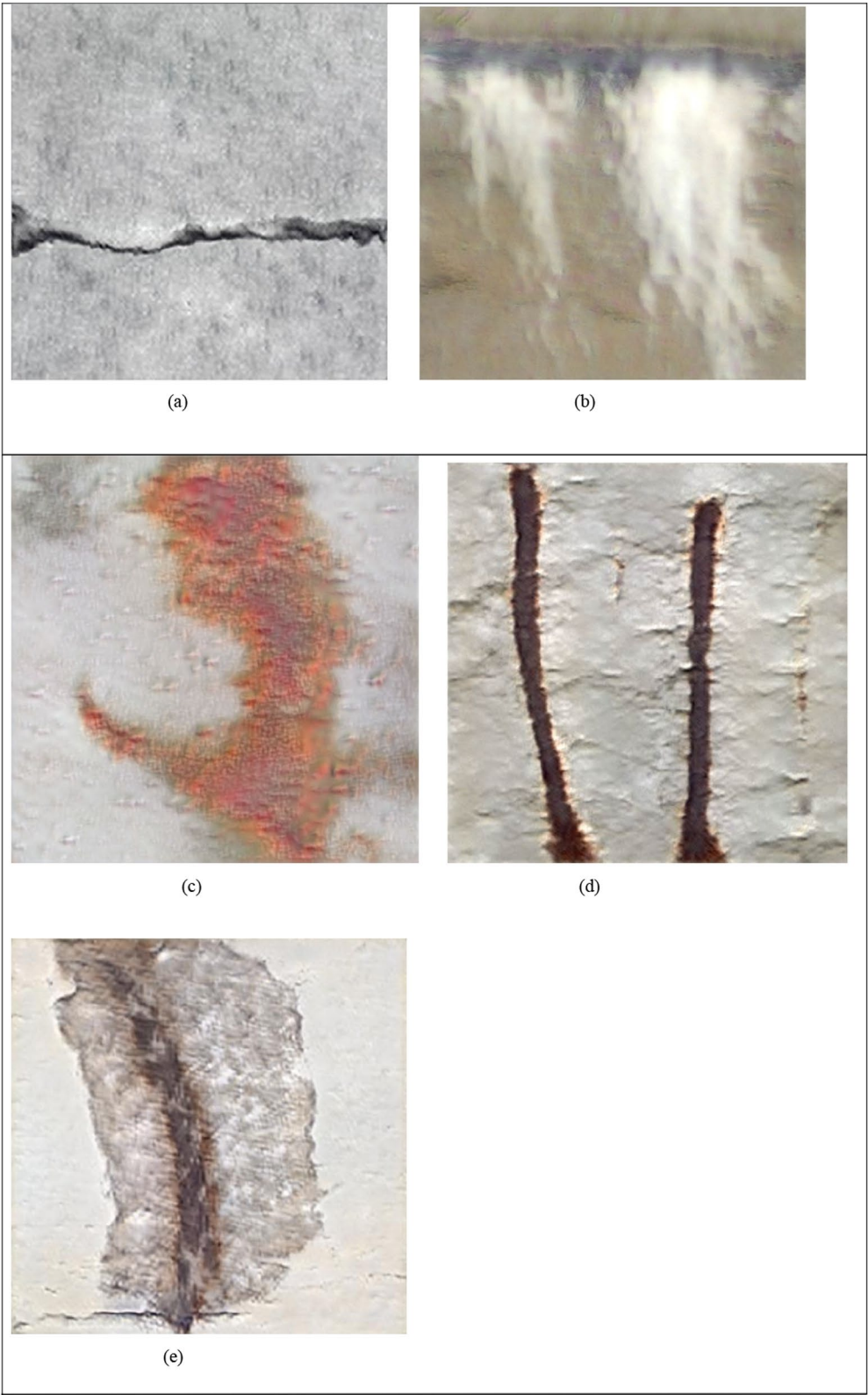


Fig. 4 Example of generated images of concrete bridge defect **a** Crack **b** Efflorescence **c** Corrosion stain **d** Exposed bars **e** Spalling

Table 3 Test set performance of pretrained models

Model	Accuracy	Precision	Recall	F1-score	Hamming loss	LRAP	Coverage error
VGG16	0.644	0.792	0.730	0.758	0.099	0.877	1.834
ResNet50	0.651	0.745	0.72	0.71	0.073	0.91	1.780
EfficientNetB0	0.632	0.721	0.75	0.77	0.08	0.82	1.830

ResNet50 established the strongest overall baseline within this constrained setting, leading in primary classification performance metrics and demonstrating more reliable ranking behavior derived from its predicted probability distributions. VGG16 and EfficientNetB0 exhibited analogous performance profiles but remained consistently inferior across both threshold-dependent and ranking-based criteria. These results identify ResNet50 as the most performant baseline under limited trainable capacity.

3.2 Fine-tuned model results

This subsection reports test-set performance for the fine-tuned configuration, in which the classification head and selected late backbone stages are unfrozen and optimized with the best hyperparameters. As shown in Table 4, fine-tuning yields substantial and consistent improvements relative to the frozen-backbone baselines across all evaluated architectures. This observation underscores the critical importance of adapting higher-level feature representations to the specific domain of bridge defect imagery for effective multi-label recognition. All models benefit from unfreezing, demonstrating uniform gains across both threshold-dependent and ranking-based metrics.

Among the fine-tuned models, VGG16 achieved the strongest overall predictive performance on the real CODEBRIM test set, leading in the primary classification metrics of Subset Accuracy (84.0%) and F1-score (85.2%). ResNet50 exhibited the most reliable probability ranking and the lowest label-wise error, as evidenced by its superior LRAP (0.942), Coverage Error (1.630), and Hamming Loss (0.057), while maintaining competitive classification accuracy. EfficientNetB0 demonstrated a compelling accuracy-efficiency tradeoff, attaining near

state-of-the-art performance with a substantially reduced model footprint and parameter count relative to both VGG16 and ResNet50. These findings indicate that moderate unfreezing substantially enhances generalization capacity, with VGG16 maximizing strict match accuracy, ResNet50 providing superior ranking quality, and EfficientNetB0 offering a compact alternative suitable for resource-constrained deployment scenarios.

3.3 Class-wise performance metrics

Table 5 summarizes class-wise accuracy, precision, recall, and F1-score on the held-out test set. Across all backbone architectures, the Crack and Exposed Bars categories achieved consistently robust performance, characterized by high F1-scores and a balanced precision-recall profile, indicative of reliable detection of these visually salient defect types. The Spalling category exhibited intermediate performance, manifesting a more discernible precision-recall tradeoff across models. This suggests that while texture-driven cues facilitate detection, they may concurrently elevate false positive rates when rough concrete surfaces are misconstrued as spalled regions.

Conversely, Efflorescence and Corrosion Stain persisted as the most challenging categories for classification. Efflorescence was characterized by high precision coupled with diminished recall, indicating conservative prediction behavior and a propensity to overlook subtle manifestations of the defect. This pattern aligns with the class's comparatively lower label support within the dataset and its visually diffuse appearance, which can be confounded with background texture variations under heterogeneous illumination. Corrosion Stain yielded the lowest F1-scores overall, reflecting a combination of moderate precision and reduced recall relative to the top-performing classes.

Table 4 Test set performance of fine-tuned models

Model	Accuracy	Epoch	Precision	Recall	F1-score	Hamming loss	LRAP	Coverage error	Size (MB)	Parameters [M]
VGG16	0.840	113	0.924	0.790	0.852	0.058	0.940	1.680	672.9	58.8
ResNet50	0.800	73	0.922	0.793	0.851	0.057	0.942	1.630	486.7	42.5
EfficientNetB0	0.823	107	0.882	0.821	0.840	0.061	0.934	1.684	227.4	19.8

Thresholded metrics (accuracy, precision, recall, F1, hamming) used a 0.5 cutoff. LRAP and coverage error were computed from raw sigmoid probabilities

Table 5 Per-class metrics for models on the test set

	Crack	Efflorescence	Exposed bars	Spalling	Corrosion stain
Accuracy					
VGG16	0.924	0.964	0.962	0.926	0.916
ResNet50	0.921	0.957	0.957	0.928	0.928
EfficientNetB0	0.905	0.961	0.955	0.921	0.915
F1 score					
VGG16	0.879	0.830	0.894	0.829	0.745
ResNet50	0.874	0.791	0.872	0.824	0.788
EfficientNetB0	0.861	0.799	0.890	0.835	0.760
Precision					
VGG16	0.894	0.912	0.941	0.924	0.841
ResNet50	0.882	0.899	0.933	0.924	0.844
EfficientNetB0	0.838	0.871	0.888	0.852	0.796
Recall					
VGG16	0.866	0.761	0.856	0.754	0.691
ResNet50	0.868	0.705	0.838	0.751	0.750
EfficientNetB0	0.870	0.767	0.876	0.811	0.745

While all three backbone architectures adhered to the same relative class difficulty ordering, they exhibited distinct error preferences. VGG16 demonstrated the most consistent per-class balance, ResNet50 exhibited superior ranking reliability and fewer label-wise errors, and EfficientNetB0 remained competitive while offering significant advantages in model compactness.

3.4 Performance on generated dataset

To assess robustness to distribution shift, the fine-tuned models were further tested on a synthetic multi-label dataset generated using StyleGAN2-ADA, as described in Sect. 2. The generated evaluation set contained $N=1,885$ images. Class frequencies, reported as label occurrences (number of images containing a positive instance), are not mutually exclusive and were distributed as follows: Crack (343), Spalling (340), Efflorescence (352), Exposed Bars (343), Corrosion Stain (350), and Background (343).

Performance metrics on the generated dataset are summarized in Table 6. Relative to the real-image CODEBRIM test set, all models exhibited diminished performance, indicating a domain shift between real and synthetic images. VGG16 achieved the highest subset accuracy on the generated set (73.2%), while ResNet50

attained the highest F1 Score (73.5%) and recall (78.2%), indicative of enhanced sensitivity to positive labels under the imposed distribution shift. EfficientNetB0 displayed the largest degradation, with lower accuracy and weaker ranking behavior, indicating less reliable label ordering on the generated images.

Class-wise results on the generated dataset are reported in Table 7. Efflorescence remained comparatively stable across models, sustaining high F1-scores, whereas Crack and Spalling yielded the lowest F1-scores, indicating that the visual cues associated with these defect categories are more challenging to recognize under the synthetic domain shift. Exposed Bars and Corrosion Stain exhibited intermediate performance, with ResNet50 demonstrating a more favorable precision-recall balance for these categories.

Based on the aggregate performance across both real and synthetic evaluation paradigms, VGG16 is designated as the primary model, attributable to its superior performance on the real CODEBRIM benchmark in terms of the primary classification metrics (Subset Accuracy and F1-score) and its robust performance on the generated dataset. ResNet50 is presented as a complementary model, distinguished by its superior recall and

Table 6 Model performance on generated dataset

Model	Accuracy	Precision	Recall	F1 score	Hamming loss	LRAP	Coverage error
VGG16	0.7321	0.7151	0.7561	0.7264	0.1075	0.8593	1.6562
ResNet50	0.7225	0.7111	0.7822	0.7345	0.1085	0.8571	1.6647
EfficientNetB0	0.6870	0.6682	0.7586	0.6900	0.1272	0.8275	1.7883

Table 7 Per-class metrics for models on the generated dataset

	Crack	Efflorescence	Exposed bars	Spalling	Corrosion stain
Accuracy					
VGG16	0.876	0.951	0.897	0.802	0.868
ResNet50	0.878	0.950	0.914	0.793	0.850
EfficientNetB0	0.868	0.930	0.869	0.797	0.837
F1 score					
VGG16	0.598	0.890	0.731	0.545	0.695
ResNet50	0.614	0.887	0.781	0.528	0.688
EfficientNetB0	0.533	0.842	0.678	0.573	0.667
Precision					
VGG16	0.733	0.908	0.698	0.455	0.624
ResNet50	0.723	0.920	0.728	0.439	0.572
EfficientNetB0	0.747	0.871	0.613	0.454	0.546
Recall					
VGG16	0.504	0.873	0.767	0.679	0.783
ResNet50	0.534	0.856	0.843	0.664	0.864
EfficientNetB0	0.414	0.815	0.758	0.776	0.858

F1-score on the synthetic evaluation set and its more reliable label-ranking characteristics on the real CODEBRIM test set.

The selected model (VGG16) was further assessed qualitatively on both the real and generated held-out test sets. Figure 5a–d shows representative multi-label predictions on real CODEBRIM test images, demonstrating the model's ability to recognize multiple co-occurring distress types within a single scene. Figure 6a–d presents representative multi-label predictions on the generated test images, providing a complementary qualitative check of model behavior in the synthetic evaluation setting.

3.5 Grad-CAM for interpretability

Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to interpret the predictions generated by the fine-tuned VGG16 model by visualizing the spatial regions within the input imagery that most strongly influence the assignment of each defect label (Selvaraju et al., 2017). For multi-label inference, Grad-CAM heatmaps were computed independently for each class by backpropagating the gradient of the target class score with respect to the feature maps of the final convolutional layer. The resultant heatmaps were superimposed onto the input images to elucidate the spatial evidence utilized by the network in its decision-making process. Representative Grad-CAM visualizations are presented in Fig. 7, revealing the model's ability to localize discriminative defect features.

Finally, Table 8 situates the performance of the proposed VGG16-based model within the context of previously reported results on the CODEBRIM dataset.

Under the evaluation protocol adopted in this study, the proposed model achieved 84.0% Subset Accuracy and an F1-score of 85.2% on the held-out test set. Relative to antecedent investigations utilizing the CODEBRIM benchmark, the proposed model demonstrates competitive, and in several instances superior, performance.

4 Discussion

This investigation systematically evaluated three ImageNet-pretrained CNN backbones for the multi-label classification of concrete bridge defects using the CODEBRIM dataset under both frozen-backbone and fine-tuned transfer-learning configurations. The frozen-backbone experiments provided a controlled baseline wherein only the classification head was trainable. Under this constraint, the modest performance differentials observed across architectures suggest that while generic ImageNet features afford useful transferability, they are inherently limited in their capacity to capture the domain-specific visual nuances of bridge defects and the complexities of label co-occurrence. The consistent and substantial performance improvements realized through hyperparameter-optimized fine-tuning (as detailed in Table 4) empirically validate the assertion that adaptation of higher-level feature representations is critical for achieving robust multi-label recognition within this application domain.

Among the fine-tuned models, VGG16 achieved the strongest performance on the real CODEBRIM test set as measured by the primary classification metrics of Subset Accuracy and F1-score, thereby justifying its selection as the primary model for reporting in-domain

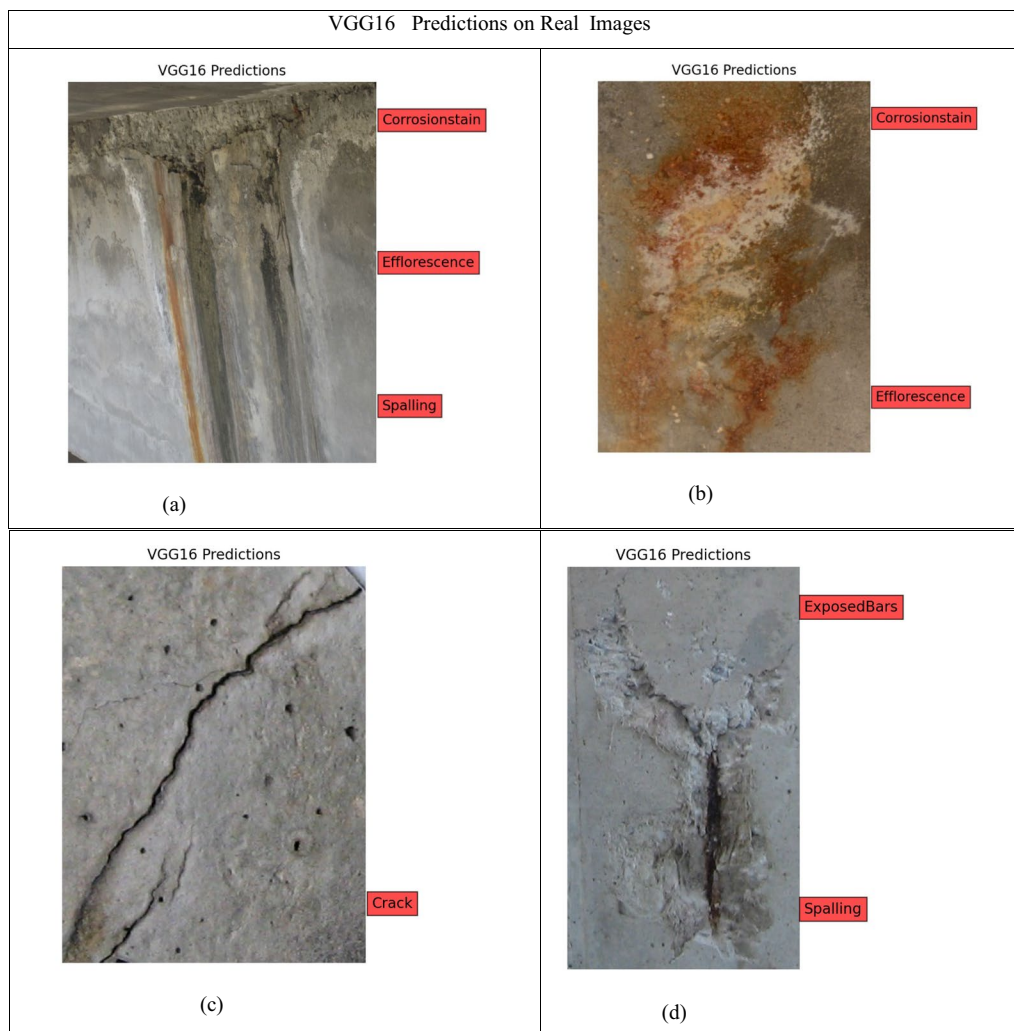


Fig. 5 VGG16 prediction results on real images. **a** Corrosion stain, efflorescence, and spalling; **b** Corrosion stain and efflorescence; **c** Crack; **d** Exposed bars and spalling

results. Conversely, ResNet50 exhibited superior label-ranking behavior on the real test set, as reflected by its higher LRAP and lower Coverage Error. EfficientNetB0 remained competitive while offering a substantially reduced model footprint relative to both VGG16 and ResNet50. These nuanced findings indicate that model selection should be guided by the specific objectives of the deployment context, balancing requirements for strict match accuracy, ranking reliability, and computational resource constraints.

Class-wise performance analysis on the real test set revealed a consistent difficulty ordering across all evaluated backbones. Crack and Exposed Bars were recognized with high precision-recall balance and elevated F1-scores, a finding consistent with the visually salient nature of distinct crack edges and exposed reinforcement. In contrast, Efflorescence and Corrosion Stain presented

greater challenges. Efflorescence was characterized by high precision but diminished recall, indicative of conservative predictions and missed positives, a behavior consistent with its relatively low label support and diffuse visual presentation under variable illumination. The lower recall observed for Corrosion Stain may also be attributed to its frequent co-occurrence with more visually dominant defects such as Spalling and Exposed Bars, wherein the model may prioritize high-contrast features, thereby suppressing the prediction of the comparatively subtle stain. These observations suggest that model performance is influenced not only by architectural capacity but also by the inherent distinctiveness of defect visual features and the effective diversity of labeled exemplars available per class.

The evaluation conducted on the StyleGAN2-ADA-generated dataset provided a complementary assessment

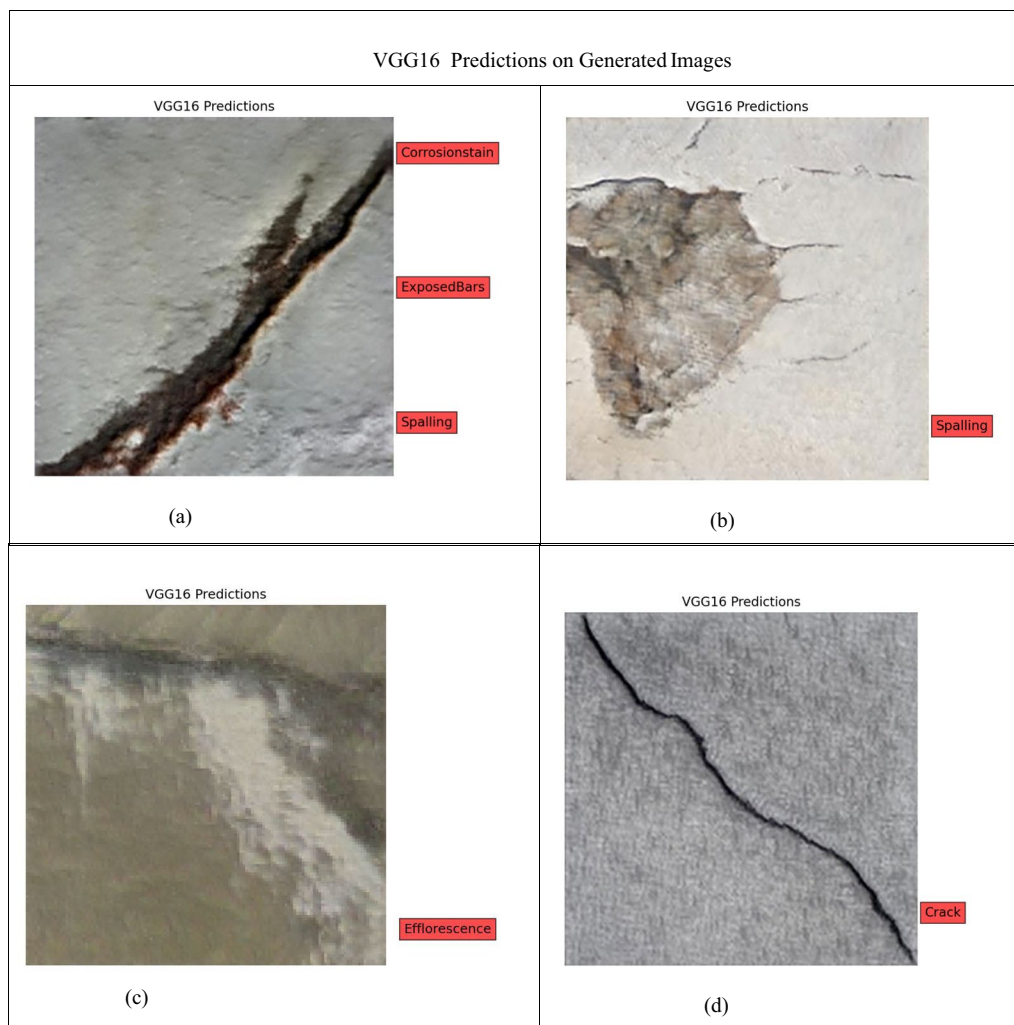


Fig. 6 VGG16 prediction results on generated images. **a** Corrosion stain, exposed bars, and spalling; **b** Spalling; **c** Efflorescence; **d** Crack

of model generalization under a real-to-synthetic domain shift (Table 6). As anticipated, all models exhibited reduced overall performance relative to the real test set, attributable to divergences in texture statistics, contrast profiles, and defect presentation characteristics. Notably, the magnitude of performance degradation varied across architectures, underscoring the importance of robustness testing for practical deployment scenarios.

The limitations of this study include the limited sample size, class imbalance, and the inherent variability of field imagery. The generated dataset is useful for robustness testing, but does not capture all real-world sources of variation. Future work should evaluate cross-bridge and cross-site generalization, incorporate class balancing, apply threshold calibration and uncertainty estimation for decision support, and extend the framework to joint localization and classification to improve defect quantification. Such advancements would facilitate the

generation of structured inspection records and support the development of probability-aware bridge digital twins for longitudinal defect tracking and optimized maintenance planning.

5 Conclusion

This study developed and rigorously evaluated a multi-label classification framework for automated concrete bridge defect recognition utilizing the CODEBRIM dataset and ImageNet-pretrained CNN backbones. Through the application of complementary multi-label evaluation metrics, the results conclusively demonstrate that hyperparameter-optimized fine-tuning yields substantial performance improvements over frozen-backbone transfer learning. This finding underscores the critical importance of domain adaptation for processing inspection imagery characterized by co-occurring defects.

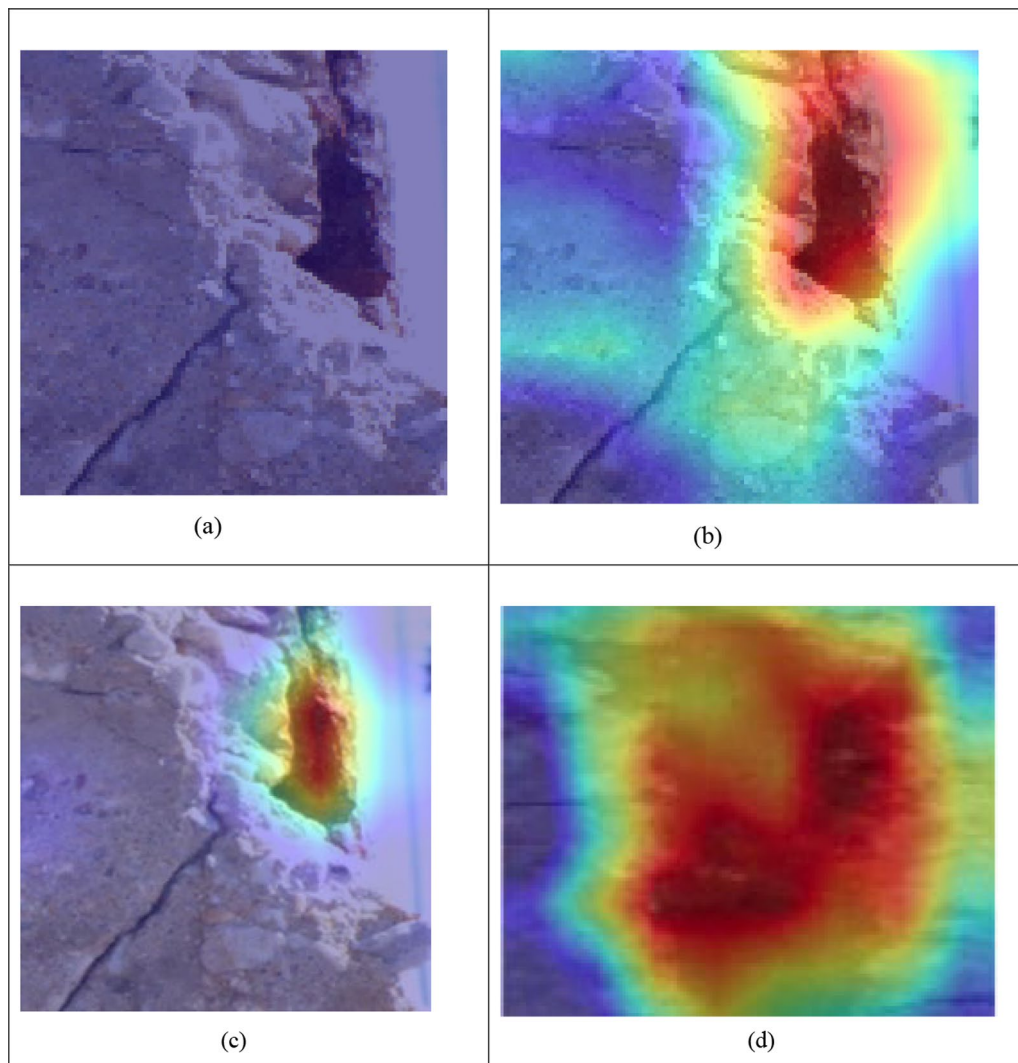


Fig. 7 Grad-CAM visualizations from VGG16. **a** Input image **b** Spalling heatmap **c** Exposed bars heatmap **d** Crack heatmap

Table 8 Performance comparison of various methods on the CODEBRIM dataset

Model	Accuracy (%)		F1 score (%)	Precision (%)	Recall (%)
	Val	Test			
(Huang & Zhang, 2024)	73.1	76.58	Not available	Not available	Not available
(Mundt et al., 2019)	64.93	72	Not available	Not available	Not available
(Che et al., 2024)	83.3	79.9	Not available	Not available	Not available
This study	81	84	85.2	92.4	79

On the real CODEBRIM test set, VGG16 achieved the strongest in-domain performance, as measured by Subset Accuracy (84.0%) and F1-score (85.2%), and was selected as the model emphasized for reporting results on the primary benchmark. ResNet50 is presented as a complementary model, providing stronger sensitivity on the

generated evaluation set and more reliable label-ranking behavior on the real test set. At the same time, EfficientNetB0 remained competitive with a smaller model footprint. Class-wise evaluation indicated that Crack and Exposed Bars were recognized most reliably, whereas Efflorescence and Corrosion Stain remained the most

challenging categories, a finding consistent with lower class support and appearance variability that increases the potential for confusion with background texture and illumination artifacts.

The proposed framework advances multi-label defect classification by combining hyperparameter-optimized fine-tuning with evaluation metrics that capture both label-wise errors and ranking quality, and by incorporating robustness and interpretability checks. Limitations include a limited dataset, class imbalance, and sensitivity to domain shift across imaging conditions. Future work will prioritize validation of generalization across diverse bridge populations and imaging conditions, and will seek to extend the framework to support defect localization and quantification, thereby enabling closer integration with operational bridge inspection workflows.

Acknowledgements

The authors gratefully acknowledge the partial support provided by the U.S. National Science Foundation.

Author contributions

Jamii Lateef: conceptualization, methodology development, data analysis, and writing. Xiong (Bill) Yu: conceptualization, supervision, and manuscript revision.

Funding

This research is partially funded by the U.S. National Science Foundation (Grant No. 2026612).

Data availability

Supporting datasets used in this study are available from the corresponding author on request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

Received: 14 May 2025 Revised: 20 February 2026 Accepted: 2 April 2026
Published online: 21 April 2026

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